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Technical Information Document

DXT Data Management System Data Interface for Donor Optimizer

The purpose of this document is to define the interface specification required for communication between DXT and external Data Management System (DMS).

This specification defines the data content of donor information and history messages passed between DXT and the DMS.

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Chapter 1 General Description

DXT is built using REST principles which ensure predictable URLs that make writing interface easy. The DXT interfaces follow HTTP rules, enabling a wide range of HTTP clients and servers to be used to interact with DXT.

This chapter gives an overview of DXT Donor Optimizer interface structure.

- Overview
- Web Services
  - Donor Information Query
  - Donor History Query
- Message Format
  - HTTP Request
  - HTTP Response
  - Root Element

Section 1.1 Overview

DXT implements bi-directional communications with external data management systems (DMS) to provide asynchronous communications. Based on the type of the message, the communications can be initiated by either DXT or the DMS. Donor Optimizer communication is always initiated by DXT as a client and DMS acts as a server.

For example, when DXT sends a donor information query to the DMS, DXT will act as a client requesting donor information from the DMS server. When the DMS responds to DXT's query, the DMS will act as a server sending donor information to the DXT client.

Section 1.2 Web Services

This section provides an overview of the web services included with DXT Donor Optimizer. The following chapter contains detailed description of each service as well as structure and content of each message.

Donor Information Query

DXT will use the Donor Information Query service to request donor information from the DMS. Donor Information is used by the DXT Donor Optimizer for platelet procedure estimation. Donor Information is used for remote procedure setup after platelet procedure estimation is complete, if DXT is licensed for the Remote Procedure Setup feature.

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Donor History Query

DXT will use the Donor History Query service to request donor history from the DMS. Donor History is used by the DXT Donor Optimizer to provide the operator with additional information during the platelet procedure estimation.

After a platelet procedure is estimated, DXT invokes remote procedure setup to send procedure parameters to the AMICUS instrument if the feature is licensed.

Section 1.3 Message Format

HTTP Request

When DXT acts as a client, it will send all HTTP request headers in the following format:

```
METHOD URL HTTP/1.1 Host: SERVER_ADDRESS
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Connection: Keep-Alive
Cache-Control: no-cache no-transform
FK-SourceId: DATA_SOURCE
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0
```

Method must be GET.

HTTP header **Host** must be present and contain network name of the destination server.

HTTP Header **Content–Length** must be present and contain payload (XML) length in bytes.

HTTP Header **Content–Type** must be “text/xml; charset=utf–8”, unless otherwise noted for the specific message.

HTTP Header **Date** must be present and contain date and time when message was generated using full Date format, specified by RFC–2616, Hypertext Transfer Protocol – HTTP/1.1. For example:

```
Date: Fri, 14 Jun 2013 21:12:59 GMT
```

DXT expects all messages to be received within 2 minutes to ensure proper functionality. DXT will respond with an error to any message that contains an HTTP header Date Time more than 2 minutes in the past or in the future from DXT server time.

HTTP Header must include **Connection: Keep–Alive** directive.

HTTP Header must include **Cache–Control** with cache directive set to “no–cache” and “no–transform”. Comma “, ” and additional values (e.g., “private”) are acceptable.

The value of HTTP Header **FK–SourceId** and **FK–DestinationId** must be set as following:

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For DXT, it should be “0”.
For DMS, it should be “1”.
For QC System, it should be “2”.

DXT expects **FK–SourceId** in HTTP Header of all inbound messages, and will supply **FK–DestinationId** in HTTP Header of all outbound messages.

For example, when the DXT sends request to DMS as a client, the IDs will be as following:

```
FK-SourceId: 0
FK-DestinationId: 1
```

Section 1.4 Root Element

All XML messages must have a root element of DMSMessage as shown below. No data is associated with this element, it is a container for all child elements.

```
<xsd:element name="DMSMessage">
    <xsd:complexType>
        <xsd:sequence>

            <CHILD ELEMENTS>

        </xsd:sequence>
    </xsd:complexType>
</xsd:element>
```

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Chapter 2 Donor Optimizer Query

This chapter defines messages that DXT sends to external data management systems for use of Donor Optimizer.

The destination URL is configurable through the DXT User Interface for each data management system DXT communicates with. Refer to DXT Data Management System Administrator's Guide Chapter 4 – DXT Configurations for details on how to configure DMS interfaces.

For possible error responses from DXT refer to Error Handling section at the end of this chapter.

Section 2.1 Donor Information Query

DXT will use the Donor Information Query service to request information about a donor from the DMS.

Donor Information Query URL

```
GET {Donor Information URL}/donor-information/{DonorId} HTTP/1.1
```

DXT will send the donor information query to the address configured as "Donor History URL" through the DXT User Interface.

{DonorId} is the unique donor identifier for whom information is being requested.

OK Response

DMS will return the following HTTP Response header when the request was processed successfully:

```
HTTP/1.1 200 OK
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-DestinationId: 0
FK-Protocol-Version: 3.0.0.0
```

The response contain the following XML Message:

```
<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <DonorInformation>
        ...
    </DonorInformation>
</DMSMessage>
```

DonorInformation XSD Definition

This complex type defines the contents of the Donor Information structure as shown below.

```
<xsd:complexType name="DonorInformation">
    <xsd:sequence>
```


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```
<xsd:element name="DonorId" >
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:minLength value="1" />
            <xsd:maxLength value="25" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="DonorGender">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:pattern value="[MF]" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="DonorHeight">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:float">
                <xsd:attribute name="units" type="xsd:boolean" use="required" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>

<xsd:element name="DonorWeight">
    <xsd:complexType>
        <xsd:simpleContent>
            <xsd:extension base="xsd:float">
                <xsd:attribute name="units" type="xsd:boolean" use="required" />
            </xsd:extension>
        </xsd:simpleContent>
    </xsd:complexType>
</xsd:element>

<xsd:element name="DonorBloodType"  minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:nonNegativeInteger">
            <xsd:minInclusive value="1" />
            <xsd:maxInclusive value="9" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="DonorPreCount" type="xsd:integer" minOccurs="0" />

<xsd:choice minOccurs="0">
    <xsd:element name="DonorHematocrit" type="xsd:integer" />
    <xsd:element name="DonorHemoglobin">
```


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```
                <xsd:complexType>
                    <xsd:simpleContent>
                        <xsd:extension base="xsd:float">
                            <xsd:attribute name="units" use="required">
                                <xsd:simpleType>
                                    <xsd:restriction base="xsd:nonNegativeInteger">
                                        <xsd:minInclusive value="1" />
                                        <xsd:maxInclusive value="2" />
                                    </xsd:restriction>
                                </xsd:simpleType>
                            </xsd:attribute>
                        </xsd:extension>
                    </xsd:simpleContent>
                </xsd:complexType>
            </xsd:element>
        </xsd:choice>

        <xsd:element name="RbcEligibility" type="xsd:boolean" minOccurs="0" />

        <xsd:element name="PlasmaEligibility" type="xsd:boolean" minOccurs="0" />

        <xsd:element name="TotalRbcLoss" type="xsd:integer" minOccurs="0" />

        <xsd:element name="TotalPlasmaLoss" type="xsd:integer" minOccurs="0" />

    </xsd:sequence>
</xsd:complexType>
```

DonorInformation Data Description

The following table explains the content of Donor Information message:

Element	Parent Element	Data Type	Description
DonorInformation	DMSMessage		Container. This tag is required.
DonorId	DonorInformation	Alphanumeric string. 25 characters max.	Donor identifier as entered by the operator in the DXT Donor Optimizer.
DonorGender	DonorInformation	String. 1 character.	Donor's gender. Acceptable values are: M (male) or F (female)
DonorHeight	DonorInformation	Float	Donor's height. A "units" attribute indicates units of measurement: false – inches true – centimeters The units must match how the DXT Donor Optimizer is configured.

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Element	Parent Element	Data Type	Description
DonorWeight	DonorInformation	Float	Donor's weight. A "units" attribute indicates units of measurement: false – pounds true – kilograms The units must match how the DXT Donor Optimizer is configured.
DonorBloodType	DonorInformation	Non-negative integer	Donor's blood type. 1 – A- 2 – A+ 3 – B- 4 – B+ 5 – AB- 6 – AB 7 – O- 8 – O+ 9 – Unknown
DonorPreCount	DonorInformation	Integer	Donor's pre-count, expressed in 1000/uL.
DonorHematocrit	DonorInformation	Integer	Donor's hematocrit, expressed in percentage. Either DonorHematocrit or DonorHemoglobin is included according to DXT Red Cell Content configuration.
DonorHemoglobin	DonorInformation	Float	Donor's hemoglobin. Either DonorHematocrit or DonorHemoglobin is included according to DXT Red Cell Content configuration. A "units" attribute indicates units of measurement: 1 – g/dL 2 – mMol/L The units must be match how the DXT Donor Optimizer is configured.
RbcEligibility	DonorInformation	Boolean	Donor's eligibility for red cell donation. false – not eligible true – eligible

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Element	Parent Element	Data Type	Description
PlasmaEligibility	DonorInformation	Boolean	Donor's eligibility for plasma donation. false – not eligible true – eligible
TotalRbcLoss	DonorInformation	Integer	Donor's total RBC loss in recent 12 months, expressed in milliliters.
TotalPlasmaLoss	DonorInformation	Integer	Donor's total plasma loss in recent 12 months, expressed in milliliters.

Donor Information Example

The OK Response contains a DonorInformation element.

```
<DonorInformation>
    <DonorId>146551591</DonorId>
    <DonorGender>F</DonorGender>
    <DonorHeight units="false">65.0</DonorHeight>
    <DonorWeight units="false">165.0</DonorWeight>
    <DonorBloodType>1</DonorBloodType>
    <DonorPreCount>250</DonorPreCount>
    <DonorHematocrit>42</DonorHematocrit>
    <RbcEligibility>true</RbcEligibility>
    <PlasmaEligibility>true</PlasmaEligibility>
    <TotalRbcLoss>100</TotalRbcLoss>
    <TotalPlasmaLoss>200</TotalPlasmaLoss>
</DonorInformation>
```

Section 2.2 Donor History Query

DXT will use the Donor History Query service to request the donation history of a donor from the DMS.

Donor History Query URL

```
GET {Donor History URL}/donor-history/{DonorId}/{StartDate}/{EndDate} HTTP/1.1
```

DXT will send donor history query to the address configured as "Donor History URL" through the DXT User Interface.

{DonorId} is the unique donor identifier for whom donation history is being requested.
{StartDate} is the first day of the donation date range for the donor history being requested, in yyyy-mm-dd format.
{EndDate} is the last day of the donation date range for the donor history being requested, in yyyy-mm-dd format.

OK Response

DMS will return the following HTTP Response header when the request was processed successfully:

```
HTTP/1.1 200 OK
```

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```
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-DestinationId: 0
FK-Protocol-Version: 3.0.0.0
```

The response contain the following XML Message:

```
<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <DonorHistory>
        ...
    </DonorHistory>
</DMSMessage>
```

DonorHistory XSD Definition

This complex type defines the contents of the Donor History structure as shown below.

```
<xsd:complexType name="DonorHistory">
    <xsd:sequence>

        <xsd:element name="DonorId">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="25" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="DonorHistoryRecord" maxOccurs="unbounded" minOccurs="0">
            <xsd:complexType>
                <xsd:sequence>

                    <xsd:element name="CollectionDate" type="xsd:date" />

                    <xsd:element name="DonationId">
                        <xsd:simpleType>
                            <xsd:restriction base="xsd:string">
                                <xsd:minLength value="1" />
                                <xsd:maxLength value="25" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:element>

                    <xsd:element name="ProcedureTypeCode"
                                 type="xsd:unsignedInt"
                                 minOccurs="0" />

                    <xsd:element name="ProcedureType"
```


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```
                type="xsd:string"
                minOccurs="0"/>

<xsd:element name="InstrumentSerialNumber" minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:minLength value="1" />
            <xsd:maxLength value="20" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="WBVolumeTarget" minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:nonNegativeInteger">
            <xsd:minInclusive value="0" />
            <xsd:maxInclusive value="99999" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="WBVolumeProcessed" minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:nonNegativeInteger">
            <xsd:minInclusive value="0" />
            <xsd:maxInclusive value="99999" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="WBProductVolume" minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:nonNegativeInteger">
            <xsd:minInclusive value="0" />
            <xsd:maxInclusive value="99999" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="TargetPreCount" minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:nonNegativeInteger">
            <xsd:minInclusive value="0" />
            <xsd:maxInclusive value="999" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="ActualPreCount" minOccurs="0" >
    <xsd:simpleType>
        <xsd:restriction base="xsd:nonNegativeInteger">
            <xsd:minInclusive value="0" />
```


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```
                <xsd:maxInclusive value="999" />
            </xsd:restriction>
        </xsd:simpleType>
    </xsd:element>

    <xsd:element name="TargetPlateletYield"
                 type="xsd:float"
                 minOccurs="0" />

    <xsd:element name="ActualPlateletYield"
                 type="xsd:float"
                 minOccurs="0" />

    <xsd:element name="StoragePlasmaVolume"
                 type="xsd:integer"
                 minOccurs="0" />

    <xsd:element name="ProcedureAlarmCount" minOccurs="0">
        <xsd:simpleType>
            <xsd:restriction base="xsd:nonNegativeInteger">
                <xsd:minInclusive value="0" />
                <xsd:maxInclusive value="999" />
            </xsd:restriction>
        </xsd:simpleType>
    </xsd:element>

    <xsd:element name="SiteId" type="xsd:string" minOccurs="0" />

    <xsd:element name="DonorHeight" minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:float">
                    <xsd:attribute name="units"
                                   type="xsd:boolean"
                                   use="required" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>

    <xsd:element name="DonorWeight" minOccurs="0">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:float">
                    <xsd:attribute name="units"
                                   type="xsd:boolean"
                                   use="required" />
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
```


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```
<xsd:choice minOccurs="0">

    <xsd:element name="DonorHematocrit">
        <xsd:simpleType>
            <xsd:restriction base="xsd:positiveInteger">
                <xsd:minInclusive value="10" />
                <xsd:maxInclusive value="99" />
            </xsd:restriction>
        </xsd:simpleType>
    </xsd:element>

    <xsd:element name="DonorHemoglobin">
        <xsd:complexType>
            <xsd:simpleContent>
                <xsd:extension base="xsd:float">
                    <xsd:attribute name="units" use="required">
                        <xsd:simpleType>
                            <xsd:restriction
                                base="xsd:nonNegativeInteger">
                                <xsd:minInclusive value="1" />
                                <xsd:maxInclusive value="2" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:attribute>
                </xsd:extension>
            </xsd:simpleContent>
        </xsd:complexType>
    </xsd:element>
</xsd:choice>

<xsd:element name="CollectionDuration"
             type="xsd:integer"
             minOccurs="0" />

<xsd:element name="PhlebotomistStartId" minOccurs="0">
    <xsd:simpleType>
        <xsd:restriction base="xsd:string">
            <xsd:minLength value="1" />
            <xsd:maxLength value="20" />
        </xsd:restriction>
    </xsd:simpleType>
</xsd:element>

<xsd:element name="PAS"
             type="xsd:boolean"
             minOccurs="0" />

<xsd:element name="StorageFluidTargetVolume"
             type="xsd:integer"
             minOccurs="0" />

<xsd:element name="ActualPlasmaProductVolume"
```


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```
                                type="xsd:integer"
                                minOccurs="0" />

                        <xsd:element name="ActualRBCProductVolume"
                                type="xsd:integer"
                                minOccurs="0" />

                    </xsd:sequence>
                </xsd:complexType>
            </xsd:element>
        </xsd:sequence>
    </xsd:complexType>
```

DonorHistory Data Description

The following table explains the content of Donor History message:

Element	Parent Element	Data Type	Description
DonorHistory	DMSMessage		Container. This tag is required.
DonorId	DonorHistory	Alphanumeric string. 25 characters max.	Donor identifier of the procedure. This element is required.
CollectionDate	DonorHistoryRecord	Date: yyyymmdd	Date when the procedure was performed. This element is required.
DonationId	DonorHistoryRecord	Alphanumeric string. 25 characters max.	Identifier of the procedure. This element is required.

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Element	Parent Element	Data Type	Description
ProcedureTypeCode	DonorHistoryRecord	Non–negative integer	Type of procedure performed[1]. Acceptable values for procedures performed on Fresenius-Kabi instruments are: 11 – Plasma, Saline (Aurora/Aurora Xi) 12 – Plasma, No Saline (Aurora/Aurora Xi) 21 – Single Needle Platelets (Amicus) 22 – Double Needle Platelets (Amicus) 31 – Mononuclear Cell (Amicus) 32 – Therapeutic Plasma Exchange (Amicus) 33 – RBC Exchange (Amicus) 41 – RBC/Plasma (Alyx) 42 – 2RBC-LR (Alyx) 43 – 2RBC-NLR (Alyx) 44 – Plasma (Alyx) 45 – RBC/Platelet (Alyx) Acceptable values for procedures performed on non-Fresenius-Kabi instruments are: 1001 – 2RBC 1002 – RBC/Plasma 1003 – Plasma 1004 – Platelet 1005 – Platelet PAS 1006 – Platelet/Plasma 1007 – Platelet PAS/Plasma 1008 – Platelet/RBC 1009 – Platelet PAS/RBC 1010 – Platelet/Plasma/RBC 1011 – Platelet PAS/Plasma/RBC 1012 – Whole Blood
ProcedureType	DonorHistoryRecord	String	Description of the procedure.
InstrumentSerialNumber	DonorHistoryRecord	String. 20 characters max.	Serial Number of the instrument that performed the procedure.
WBVolumeTarget	DonorHistoryRecord	Non–negative integer. 0 – 99999 range.	Target whole blood volume of the procedure, expressed in milliliters.
WBVolumeProcessed	DonorHistoryRecord	Non-negative integer. 0 – 99999 range.	Volume of whole blood processed during the procedure, expressed in milliliters.
WBProductVolume	DonorHistoryRecord	Non-negative integer. 0 – 99999 range.	Actual whole blood volume of the procedure, expressed in milliliters.

[1] If DMS include procedure type, DXT displays the procedure type. If DMS does not include procedure type, DXT maps the procedure type code to a procedure type. DXT displays an unknown procedure type code (e.g. “unknow (9999”)) or displays an empty string if both procedure type and procedure type code are not included.

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Element	Parent Element	Data Type	Description
TargetPreCount	DonorHistoryRecord	Non-negative integer. 0 – 999 range.	Target concentration of platelets, expressed in 1000/uL.
ActualPreCount	DonorHistoryRecord	Non-negative integer. 0 – 999 range.	Actual concentration of platelets, expressed in 1000/uL.
TargetPlateletYield	DonorHistoryRecord	Float	Target platelet yield in the product, expressed in xe11
ActualPlateletYield	DonorHistoryRecord	Float	Actual platelet yield in the product, expressed in xe11
StoragePlasmaVolume	DonorHistoryRecord	Integer	Actual storage fluid volume for the platelet product, expressed in milliliters. This volume includes PAS if PAS is used.
ProcedureAlarmCount	DonorHistoryRecord	Non-negative integer. 0 – 999 range.	Total number of alarms occurred during the procedure.
SiteId	DonorHistoryRecord	String	Site identifier where the procedure was performed.
DonorHeight	DonorHistoryRecord	Float	Donor's height. A "units" attribute indicates units of measurement: false – inches true – centimeters The units must match how the DXT Donor Optimizer is configured.
DonorWeight	DonorHistoryRecord	Float	Donor's weight. A "units" attribute indicates units of measurement: false – pounds true – kilograms The units must match how the DXT Donor Optimizer is configured.
DonorHematocrit	DonorHistoryRecord	Integer	Donor's hematocrit, expressed in percentage. Either DonorHematocrit or DonorHemoglobin is included according to DXT Red Cell Content configuration.

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Element	Parent Element	Data Type	Description
DonorHemoglobin	DonorHistoryRecord	Float	Donor's hemoglobin. Either DonorHematocrit or DonorHemoglobin is included according to DXT Red Cell Content configuration. A "units" attribute indicates units of measurement: 1 – g/dL 2 – mMol/L The units must be match how the DXT Donor Optimizer is configured.
CollectionDuration	DonorHistoryRecord	Integer	Collection time during the procedure, expressed in minutes.
PhlebotomistStartId	DonorHistoryRecord	String. 20 characters max.	The identifier of the phlebotomist that performed the venipuncture.
PAS	DonorHistoryRecord	Boolean	Indication whether PAS was used in the procedure. false – PAS was not used true – PAS was used
StorageFluidTargetVolume	DonorHistoryRecord	Integer	Targeted storage fluid volume for the platelet product, expressed in milliliters. This volume includes PAS if PAS was used.
ActualPlasmaProductVolume	DonorHistoryRecord	Integer	Actual plasma volume of the procedure, expressed in milliliters.
ActualRBCProductVolume	DonorHistoryRecord	Integer	Actual RBC volume of the procedure, expressed in milliliters.

Donor History Example

The OK Response contains one DonorHistory element and any number DonorHistoryRecord elements.

```
<DonorHistory>
    <DonorId>146551591</DonorId>
    <DonorHistoryRecord>
        <CollectionDate>2019-03-02</CollectionDate>
        <DonationId>AS0066</DonationId>
        <ProcedureTypeCode>21</ProcedureTypeCode>
        <InstrumentSerialNumber>AMI6470</InstrumentSerialNumber>
        <WBVolumeTarget>3031</WBVolumeTarget>
        <WBVolumeProcessed>6521</WBVolumeProcessed>
        <WBProductVolume>3597</WBProductVolume>
        <TargetPreCount>357</TargetPreCount>
        <ActualPreCount>437</ActualPreCount>
        <TargetPlateletYield>7.638843</TargetPlateletYield>
        <ActualPlateletYield>3.81053448</ActualPlateletYield>
        <StoragePlasmaVolume>409</StoragePlasmaVolume>
        <ProcedureAlarmCount>0</ProcedureAlarmCount>
```


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```
        <SiteId>Lake</SiteId>
        <DonorHeight units="false">65.0</DonorHeight>
        <DonorWeight units="false">165.0</DonorWeight>
        <DonorHematocrit>42</DonorHematocrit>
        <CollectionDuration>116</CollectionDuration>
        <PhlebotomistStartId>Phleb_9153</PhlebotomistStartId>
        <PAS>true</PAS>
        <StorageFluidTargetVolume>373</StorageFluidTargetVolume>
        <ActualPlasmaProductVolume>395</ActualPlasmaProductVolume>
        <ActualRBCProductVolume>0</ActualRBCProductVolume>
    </DonorHistoryRecord>
</DonorHistory>
```

Section 2.3 DMS Error Response

Donor Not Found Error Response

When DMS cannot identify the donor in the request, the following response will be returned to the sender:

```
HTTP/1.1 500 Internal Server Error
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-DestinationId: 0
FK-Protocol-Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
        <Code>ERROR_CODE</Code>
        <Description>ERROR_DESCRIPTION</Description>
    </ErrorResponse>
</DXTMessage>
```

The following table lists the Error code and descriptions for the error:

Error Code	Error_Description	Notes
50017	Donor Id Not Found	DonorId element in DXT's Donor Information Query or Donor History Query request is not a valid donor identifier in the DMS.

Request Validation Error Response

When DMS encounters problems validating the request, the following response will be returned to the DXT:

```
HTTP/1.1 400 Bad Request
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-DestinationId: 0
FK-Protocol-Version: 3.0.0.0
```


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```
<DXTMessage>
    <ErrorResponse>
        <Code>ERROR_CODE</Code>
        <Description>ERROR_DESCRIPTION</Description>
    </ErrorResponse>
</DXTMessage>
```

DXT does not specifically act on these types of error responses.

Over time, this information may be subject to change based upon modification in product manufacturing, blood banking, or critical practice. For additional questions please contact the Clinical Technical Hotline at 1–800–937–5060.

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